

Making KataGo HumanSL More Human-Like for Amateur-Level Play [★]

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Abstract. Imitating human gameplay is one of the important factors in entertaining or teaching human players. In this paper, we target the game of Go at amateur levels and investigate KataGo’s HumanSL model, which was trained on a large number of human game records using supervised learning (SL). Specifically, we (1) evaluate the human-likeness of HumanSL policies (i.e., probability distribution of moves) at various skill levels from several aspects, (2) propose a method called *Policy Adjustment based on Previous Move (PAPM)* to enhance the human-likeness of HumanSL policies, and (3) demonstrate that selecting moves using a *softmax* function with appropriate parameter settings can better fit the specified skill levels. Additionally, we conduct a preliminary experimental study to compare the probability distributions of HumanSL and PAPM with softmax (*PAPM+SX*). The results indicate that the participants found PAPM+SX to be more human-like than HumanSL.

Keywords: Go · KataGo HumanSL · Human-likeness · Move imitation.

1 Introduction

Reproducing human players’ behavior is important for entertaining and teaching human players. Such AI agents can serve as practice opponents or test players during content generation. In this study, we focus on making average-style moves that resemble those made by players at a given skill level. We leave as future work to reproduce other aspects of behaviors, including tactical preferences, personal traits, and non-move behaviors such as thinking time and resignation timing.

A promising approach to generating human-like moves is learning from human gameplay data. In board games, supervised learning using human game records is commonly employed. While earlier studies aimed to build strong AI agents by training on expert-level data [1,10], recent works have increasingly focused on modeling amateur players [3,6,8,11].

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In the game of Go, a well-known strong AI agent named KataGo is available. In addition to the neural networks trained via self-play, the v15 version [3] includes a neural network trained on a large number of human game records, referred to as the HumanSL model. However, it remains unclear how human-like HumanSL’s policies (i.e., the selection probabilities of moves) are, and there is no established consensus on how to use the policies to generate human-like moves.

In this study, we first examine the human-likeness of HumanSL policies at various skill levels using four metrics. While human-likeness is often measured by move-matching accuracy (i.e., the proportion of states where the move with the highest predicted probability matches the actual human move), it loses the information about the overall distribution. For example, predictions of 90% vs 10% and 51% vs 49% yield the same accuracy yet differ in plausibility; if humans’ gameplay is closer to 90% vs 10%, the former is preferable, and *vice versa*. To address this limitation, we also employ likelihood [7], skill level, and average distance to the previous move as complementary metrics.

Furthermore, we propose *Policy Adjustment based on Previous Move (PAPM)* to improve human-likeness by favoring moves closer to the previous one. We expect PAPM to be particularly effective in Go, but we believe that the tendency of weaker players to focus on recent moves is applicable to other games. Finally, regarding the use of the predicted move probability distributions, previous studies typically select the most probable move or sample according to the distribution (e.g., [3,6,10]). We show that a *softmax* interpolation with appropriate parameter settings can better fit the specified skill levels.

2 Related Work

McIlroy-Young et al. [6] proposed *Maia*, a collection of models trained on amateur chess game records, with separate supervised learning (SL) models for different skill levels. Move-matching accuracy was highest when training and testing skill levels were similar. Although the use of the predicted move probability distributions was not explicitly described, the metric choice and the released code¹ imply selection of the move with the highest predicted probability. They later introduced *Maia-2*, a unified model with skill-aware attention for predicting human moves across various skill levels [11].

KataGo’s HumanSL is similar to Maia-2 in using a single model for a wide range of skill levels [3], including amateurs and professionals. It takes as input not only the board state but also metadata such as the skill levels for both players and the game date. It then outputs a move probability distribution along with other predictions such as win rates. KataGo’s API allows one to specify the skill level of amateurs using settings `preaz_20k`, ..., `preaz_1k`, `preaz_1d`, ..., `preaz_9d` or `rank_20k`, ..., `rank_1k`, `rank_1d`, ..., `rank_9d`. `preaz` and `rank` differ in that the former reflects the pre-AlphaZero opening style, while the latter reflects

¹ <https://github.com/CSSLab/maia-chess/blob/e464f161c295f4246c2082f7b36d646d9bd14e7c/README.md?plain=1#L60>

the modern opening style. The documentation² recommends sampling moves according to the predicted distribution, which differs from Maia’s implication of selecting moves with the highest probability.

3 Metrics of Human-Likeness and Preliminary Results

In this section, we evaluate how well policies predicted by KataGo’s HumanSL align with the specified skill levels of amateur Go players, using four evaluation metrics introduced in Subsections 3.1 to 3.4. In Go, *rank* (dan/kyu) is commonly used to represent skill levels. For example, on an online Go server, Fox Go, the rank ranges from 18k (weakest) to 9d (strongest).³ In our experiments, we used 1,000 game records per rank from a publicly available dataset⁴ of games played on Fox Go, sampling only those where both players had the same rank. Since most of those games were played before AlphaZero-style openings became popular, we used the **preaz** settings for HumanSL. For comparison, we also tested KataGo’s self-play model, referred to as **kata1**.

3.1 Move-Matching Accuracy

As a basic metric of human-likeness, we first considered the commonly used move-matching accuracy, i.e., the proportion of states where the move with the highest predicted probability matches the actual move in the game. Fig. 1a shows the accuracy of HumanSL with various **preaz** settings (rainbow-colored curves) on the collected games from Fox Go at each rank from 18k to 9d (x-axis). For clarity, results for HumanSL are plotted only for every few ranks (19k, 16k, ..., 9d). For each rank, the highest accuracy was generally obtained with the corresponding or a nearby **preaz** setting, consistent with previous work [6,7]. As a reference, the accuracy of **kata1** (dotted black curve) was lower than that of HumanSL, indicating it is relatively less human-like.

3.2 Likelihood

Ogawa et al. [7] proposed using *likelihood* to evaluate human-likeness in shogi, defined as the geometric mean of the predicted probabilities for the human player’s moves. Given a set of state-move pairs $\{(s_i, a_i) | i \in [1..N]\}$ in human

² https://github.com/lightvector/KataGo/blob/802946dbb15ab7b52f6fa18e777ec8f8f65bfaff/docs/Analysis_Engine.md?plain=1#L464

³ On different Go servers, the same rank values may correspond to different skill levels. For example, 1d on Fox Go was reported to correspond to 2k on KGS (see <https://senseis.xmp.net/?RankWorldwideComparison>). **preaz** and **rank** settings of HumanSL are based on KGS (see <https://github.com/lightvector/KataGo/blob/802946dbb15ab7b52f6fa18e777ec8f8f65bfaff/cpp/neuralnet/sgfmetadata.cpp#L230>). We ignored such deviations in this work to make the results easier to understand.

⁴ <https://github.com/featurecat/go-dataset>

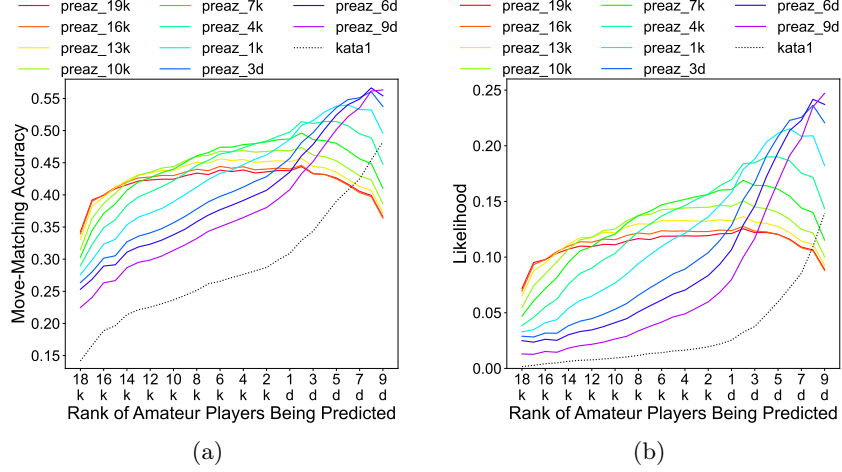


Fig. 1: (a) Move-matching accuracy and (b) likelihood of KataGo’s HumanSL `preaz_{19k, 16k, ..., 9d}` and `kata1` in the game records used in the experiments that were from the Fox Go Dataset.

game records, the likelihood is $(\prod_{i=1}^N P(s_i, a_i))^{(1/N)}$, where $P(s, a)$ is the model’s predicted probability of playing move a at state s .

Unlike move-matching accuracy, likelihood focuses on the probability assigned to the actual human move, offering a more fine-grained view of the model’s predictions. For example, if a state s with two moves a_1 and a_2 has the true distribution of 80% and 20%, a model predicting 80-20 and another predicting 60-40 yield the same accuracy (80%) but different likelihoods (0.61 versus 0.55), reflecting confidence alignment. In particular, likelihood penalizes the model more heavily when the actual move is assigned a low probability.

Fig. 1b shows likelihood results for HumanSL on Fox Go games from 18k to 9d (x-axis). Similar to Fig. 1a, the highest likelihood was achieved with the corresponding or nearby `preaz` setting. However, the curve shapes differ, particularly for stronger `preaz` settings. Such settings often assign extremely low probabilities to weak human players’ moves, treating these moves as impossible to play at high levels. These results highlight that likelihood can capture slight confidence gaps between skill levels, complementing move-matching accuracy.

3.3 Skill Level

Ideally, the skill level of HumanSL at each setting should match and be evaluated through playing against a wide range of human players. However, collecting sufficient human-versus-AI matches is difficult. Instead, we employed as baselines Go engines with known skill levels: GNU Go (level 10)⁵, estimated at about

⁵ <https://www.gnu.org/software/gnugo/>

kgs8k and Pachi⁶ configured as kgs3k and kgs2d. We paired these with HumanSL settings of `preaz_8k`, `preaz_3k`, and `preaz_2d`, respectively. For each setting, we tested two move-selection methods: (1) greedy, always selecting the move with the highest predicted probability, and (2) sampling from the predicted probability distribution. Each matchup consisted of 500 games on a 19×19 board under Japanese rules with 6.5 komi, alternating playing Black and White.

The greedy version of HumanSL won far above the expected skill levels: with `preaz_8k`, it had a win rate with a 95% confidence interval of $82.6 \pm 3.3\%$ against GNU Go (kgs8k); with `preaz_3k`, it had $97.8 \pm 1.3\%$ against Pachi_{kgs3k}; and with `preaz_2d`, it had $94.6 \pm 2.0\%$ against Pachi_{kgs2d}. Conversely, the sampling version performed much weaker than the expected skill levels: $26.4 \pm 3.9\%$, $36.6 \pm 4.2\%$, and $23.4 \pm 3.7\%$ for the same matchups, respectively. A similar gap has been observed in chess for models trained on amateur data [12].

We hypothesize two main causes: First, greedy selection works like a majority vote. For example, if 6 out of 10 players choose a winning move A and 4 choose a losing move B in a given state, and a well-learned model predicts 60% for A and 40% for B, greedy selection always chooses A, making the model stronger than the average player. Second, although sampling would match the average player under this scenario, the assumption of well-learned models rarely holds in practice. Even for obviously bad moves that weak players can avoid with minimal reading, the models may still assign non-zero probabilities. Sampling can then select such moves, leading to weaker play. These findings suggest that alternative ways of using the predicted move probability distribution may better fit the intended skill level.

3.4 Distance to the Previous Move

In Go, human players tend to respond near the opponent’s last move, and this tendency is believed to be stronger among weaker players. To quantify this behavior, we measure the distance between the current move and the previous move, using a metric proposed by Rémi Coulom [1], hereafter referred to as the Rémi distance. It is defined as $|\delta_x| + |\delta_y| + \max(|\delta_x|, |\delta_y|)$, where δ_x and δ_y are the differences in x- and y-coordinates between two moves. This measure is argued to better fit human intuition than Euclidean or Manhattan distance in Go.

Fig. 2 shows the average Rémi distance between each move and the previous move. Brown dots indicate human data from Fox Go games, showing the expected tendency (shorter distances for weaker ranks, except 18k). Rainbow curves show HumanSL settings with sampling selection, while greedy selection (not shown in the figure) gave slightly shorter distances. Nevertheless, both deviated from human data, suggesting room for improvement. As a reference, `kata1` (dotted black curve) generally shows longer distances, indicating a weaker local response tendency.

⁶ <https://github.com/pasky/pachi>

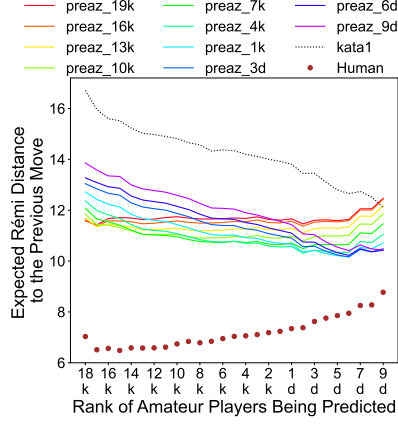


Fig. 2: Rémi distance to the previous move of KataGo’s HumanSL `preaz_{19k, 16k, ..., 9d}` and `kata1` (curves), along with Rémi distance of human players’ moves to the previous moves (dots) in the game records used in the experiments that were from the Fox Go Dataset.

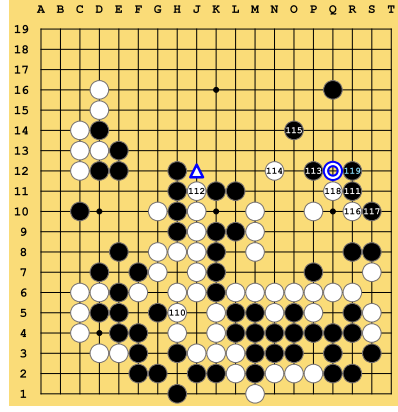


Fig. 3: A state where a 1k human player’s move matches the move with the maximum selection probability of PAPM (circle: the actually played move, $0.077 \rightarrow 0.413$; triangle: the move with the maximum selection probability of `preaz_1k` without the adjustment, $0.720 \rightarrow 0.241$)

4 Policy Adjustment Based on Previous Move (PAPM)

As shown in Subsection 3.4, HumanSL policies (i.e., move selection probabilities) tend to play farther from the previous move than human players. We address this by proposing *Policy Adjustment based on Previous Move (PAPM)*. We then evaluate the adjusted policies in Subsection 4.2.

4.1 Approach

To encourage moves closer to the previous move and discourage distant ones, we reweight HumanSL’s predicted selection probabilities $P(s, a)$ as $P'(s, a) = (W(a^-, a) \times P(s, a)) / (\sum_b (W(a^-, b) \times P(s, b)))$, where a^- is the previous move. Let d be the Rémi distance between a^- and a (if neither is a pass); the weight is

$$W(a^-, a) = \begin{cases} 1, & \text{if } a^- \text{ or } a \text{ is a pass} \\ 0.1, & \text{if } d > 14 \\ 0.1 + (14 - d) \times m, & \text{if } 14 \geq d > 4 \\ 0.1 + (14 - 4) \times m, & \text{if } 4 \geq d \end{cases} \quad (1)$$

with slope m controlling the degree of adjustment. Larger values of m increase the tendency to play near the previous move. This weighting method follows Shi

Table 1: The Rémi distance of human players’ moves to the previous moves and the Rémi distance to the previous move of HumanSL when selecting moves according to the probability distributions, with and without PAPM, in the target game records from the Fox Go Dataset. For HumanSL, bold values are those closest to human statistics.

Game Records	Human	HumanSL							
		Rank Setting	W/O PAPM	Slope Parameter m					
				0.03	0.05	0.1	0.15	0.2	0.25
16k	6.56	preaz_16k	11.58	8.94	8.28	7.42	6.97	6.67	6.46
13k	6.58	preaz_13k	11.22	8.77	8.15	7.34	6.91	6.63	6.43
10k	6.74	preaz_10k	11.04	8.76	8.18	7.41	6.99	6.72	6.53
7k	6.84	preaz_7k	10.74	8.65	8.11	7.39	7.00	6.75	6.57
4k	7.06	preaz_4k	10.70	8.73	8.21	7.52	7.15	6.90	6.72
1k	7.24	preaz_1k	10.70	8.82	8.33	7.66	7.30	7.06	6.88
3d	7.62	preaz_3d	10.58	8.82	8.35	7.71	7.36	7.13	6.96
6d	7.95	preaz_6d	10.22	8.56	8.13	7.54	7.22	7.00	6.84
9d	8.77	preaz_9d	10.44	8.75	8.31	7.71	7.39	7.17	7.01

et al. [9]⁷ but replaces the Euclidean distance with Rémi distance to better align with human players’ perspectives. Additionally, the step function is replaced with a linear function, as seen in the third line of Eq. (1). Although this change ultimately results in another step function because Rémi distances are integers, it helps reduce the number of parameters.

4.2 Evaluation Results

Distance to the Previous Move We first evaluated the effect of PAPM on the distance to the previous move. Among the game records from Fox Go used in Section 3, we selected nine skill levels for further investigation: 16k, 13k, 10k, 7k, 4k, 1k, 3d, 6d, and 9d. The second column of Table 1 shows the average Rémi distances of human moves (brown dots in Fig. 2). For HumanSL, we evaluated the **preaz** setting corresponding to the rank of each game. The fourth column shows the average Rémi distance when moves were sampled from HumanSL’s original probability distributions, i.e., without adjustment. As shown in Fig. 2, these values deviate from the human statistics.

For PAPM, we tested slope parameters $m = 0.03, 0.05, 0.1, 0.15, 0.2$, and 0.25 in Eq. (1) (columns 5 to 10 in Table 1) and obtained the average Rémi distance when moves were sampled from the PAPM distributions. With appropriate m , distances approached human statistics. In addition, lower ranks (weaker players) required larger m , indicating a stronger tendency to respond locally, as also observed in Subsection 3.4.

⁷ Let d^E be the Euclidean distance between a move to the previous move, the weights were 1.50 for $d^E \leq 2$, 1.25 for $2 < d^E \leq 3$, 1.00 for $3 < d^E \leq 4$, 0.75 for $4 < d^E \leq 5$, 0.50 for $5 < d^E \leq 6$, 0.25 for $6 < d^E \leq 7$, and 0.10 for $d^E > 7$.

Table 2: The move-matching accuracy and likelihood with and without PAPM.

Game Records	Rank Setting	Move-Matching Accuracy			Likelihood		
		W/O	W/	Improv.	W/O	W/	Improv.
16k	preaz_16k	0.400	0.468	+0.068	0.098	0.130	$\times 1.333$
13k	preaz_13k	0.431	0.494	+0.063	0.117	0.154	$\times 1.308$
10k	preaz_10k	0.444	0.502	+0.058	0.125	0.160	$\times 1.277$
7k	preaz_7k	0.466	0.518	+0.052	0.141	0.175	$\times 1.238$
4k	preaz_4k	0.473	0.522	+0.048	0.146	0.178	$\times 1.220$
1k	preaz_1k	0.474	0.519	+0.045	0.147	0.177	$\times 1.208$
3d	preaz_3d	0.497	0.534	+0.037	0.167	0.194	$\times 1.161$
6d	preaz_6d	0.540	0.565	+0.025	0.214	0.235	$\times 1.098$
9d	preaz_9d	0.563	0.578	+0.015	0.247	0.259	$\times 1.047$

Move-Matching Accuracy and Likelihood We next examined the effect of PAPM on move-matching accuracy and likelihood. For each skill level, we chose the m value that produced the closest Rémi distance to human statistics: $m = 0.25$ for **preaz_16k**, $m = 0.2$ for **preaz_13k**, **preaz_10k**, and **preaz_7k**, $m = 0.15$ for **preaz_4k** and **preaz_1k**, $m = 0.1$ for **preaz_3d**, $m = 0.05$ for **preaz_6d**, and $m = 0.03$ for **preaz_9d**. The results of move-matching accuracy and likelihood with and without PAPM are shown in Table 2. In all cases, both metrics improved after applying the adjustment, with larger improvements observed for weaker players.

To illustrate the improvement mechanism, Fig. 3 shows an example from a game between two 1k players on Fox Go. Black played at R12 for move 119, followed by White’s Q12 for move 120. Since the preceding fight was on the right side, directly responding at Q12 appears natural for kyu-level players. However, with the **preaz_1k** setting, the selection probability for Q12 at move 120 was only 0.077, much lower than 0.720 for J12, a strong penetration locally. After applying PAPM with $m = 0.15$, the probabilities for Q12 and J12 became 0.413 and 0.241, respectively, making Q12 the most probable move and matching the human move. In other words, the actual move was no longer underestimated at 0.077 but was instead predicted to have a high probability of 0.413. Similar improvements occurred in many other states, contributing to better distance alignment and higher move-matching accuracy and likelihood.

Skill Level Finally, we evaluated the effect of PAPM on skill level, as in Subsection 3.3. Players using **preaz_8k** with $m = 0.2$, **preaz_3k** with $m = 0.15$, and **preaz_2d** with $m = 0.1$ were matched against GNU Go, Pachi_{kgs3k}, and Pachi_{kgs2d}, respectively. Both greedy and sampling selection methods were tested. Each matchup consisted of 500 games with alternating colors.

For greedy selection, the win rates along with 95% confidence intervals were $74.4 \pm 3.8\%$ for **preaz_8k** against GNU Go, $89.6 \pm 2.7\%$ for **preaz_3k** against Pachi_{kgs3k}, and $85.8 \pm 3.1\%$ for **preaz_2d** against Pachi_{kgs2d}. For sampling selection, the win rates were $31.0 \pm 4.1\%$, $45.0 \pm 4.4\%$, and $29.8 \pm 4.0\%$, respectively.

Similar to Subsection 3.3, greedy players were stronger than expected, while sampling players were slightly weaker. Although all win rates approached 50% compared to those in Subsection 3.3, there is still room for improvement.

5 Softmax Selection of Policy

In Subsections 3.3 and 4.2, we found that selecting the move with the highest predicted probability tends to be overly strong, whereas sampling moves strictly according to the predicted distribution results in being slightly weaker. In this section, we target a balanced skill level using an intermediate softmax-based approach, explained in Subsection 5.1. We then present quantitative and qualitative evaluation results in Subsections 5.2 and 5.3, respectively.

5.1 Approach

Let $P(s, a)$ denote the predicted probability of move a at state s . Selecting the move with the highest predicted probability can be expressed as $P'(s, a) \propto (P(s, a))^\infty$, precisely, $P'(s, a) = \lim_{k \rightarrow \infty} (P(s, a)^k / \sum_{b \in A(s)} P(s, b)^k)$, where $A(s)$ is a set consisting of legal moves at s . Sampling moves directly from the original distribution corresponds to $P'(s, a) \propto (P(s, a))^1$. An intermediate policy can be obtained by $P'(s, a) \propto (P(s, a))^\rho$ subject to $\rho \in (1, \infty)$. The skill levels increase as ρ becomes larger, as will be shown in Subsection 5.2. This approach is referred to as *softmax* in prior studies [2,5].

5.2 Quantitative Evaluation

Following the same procedure as in Subsections 3.3 and 4.2, we evaluated PAPM with softmax (*PAPM+SX*) created from **preaz_8k**, **preaz_3k**, and **preaz_2d**, which played against GNU Go, Pachi_{kgs3k}, and Pachi_{kgs2d}, respectively. The same PAPM parameters as in Subsection 4.2 were used (i.e., $m = 0.2$ for **preaz_8k**, $m = 0.15$ for **preaz_3k**, and $m = 0.1$ for **preaz_2d**). For each **preaz** setting, we tested softmax parameters $\rho \in \{2.00, 1.50, 1.20, 1.10, 1.05\}$.

The win rates and 95% confidence intervals are shown in Table 3. For each **preaz** setting, an appropriate choice of ρ achieved a win rate close to 50%, indicating the possibility of tuning skill level. The corresponding values of ρ were 1.20, 1.05, and 1.20 for **preaz_8k**, **preaz_3k**, and **preaz_2d**, respectively. No consistent pattern for optimal ρ was found across settings.

We examined the other three metrics as well. *PAPM+SX* yielded the same move-matching accuracies as PAPM (0.516, 0.522, 0.532 for **preaz_8k**, **preaz_3k**, and **preaz_2d**) since applying softmax does not affect the top move, outperforming HumanSL's (0.462, 0.475, 0.491). The likelihoods (0.161, 0.176, 0.173) were slightly lower than those of sampling from PAPM (0.173, 0.179, 0.190) but still higher than HumanSL (0.138, 0.147, 0.161). The average Rémi distances to the previous move were reduced to 6.332, 7.115, and 7.316 from 6.700, 7.214, and 7.608, substantially lower than HumanSL's 10.799, 10.741, and 10.487. Overall, softmax selection did not have too much negative influence on human-likeness.

Table 3: Win rates and the 95% confidence intervals for players based on KataGo’s HumanSL with PAMP+**SX** playing against GNU Go, Pachi_{kgs3k}, and Pachi_{kgs2d}. *Italic values are those closest to 50%.*

Opponent	GNU Go	Pachi _{kgs3k}	Pachi _{kgs2d}
HumanSL	preaz_8k	preaz_3k	preaz_2d
PAMP	$m = 0.2$	$m = 0.15$	$m = 0.1$
$\rho = \infty$	74.40 $\pm$ 3.83%	89.60 $\pm$ 2.68%	85.80 $\pm$ 3.06%
$\rho = 2.00$	66.60 $\pm$ 4.13%	83.20 $\pm$ 3.28%	74.40 $\pm$ 3.83%
$\rho = 1.50$	55.20 $\pm$ 4.36%	77.40 $\pm$ 3.67%	67.60 $\pm$ 4.10%
$\rho = 1.20$	<i>50.20$\pm$4.38%</i>	68.40 $\pm$ 4.08%	<i>47.80$\pm$4.38%</i>
$\rho = 1.10$	40.80 $\pm$ 4.31%	59.20 $\pm$ 4.31%	40.40 $\pm$ 4.30%
$\rho = 1.05$	39.60 $\pm$ 4.29%	<i>51.80$\pm$4.38%</i>	37.80 $\pm$ 4.25%
$\rho = 1.00$	31.00 $\pm$ 4.05%	45.00 $\pm$ 4.36%	29.80 $\pm$ 4.01%

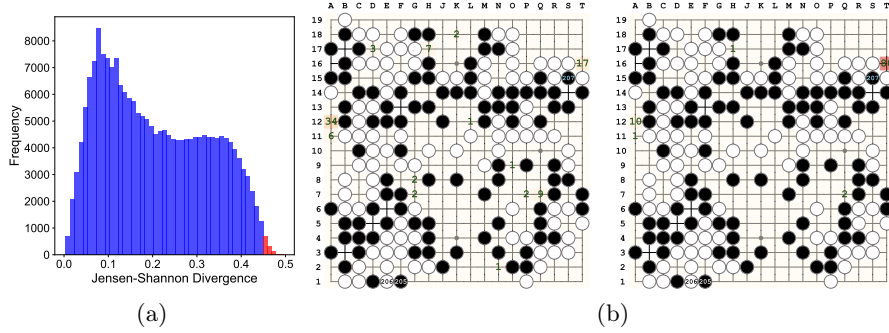


Fig. 4: (a) the histogram of Jensen-Shannon divergence of PAMP+**SX** and the original **preaz_2d** in the collected Fox 2d games, and (b) an example pair of policies from **preaz_2d** and PAMP+**SX**.

5.3 Qualitative Evaluation

We conducted a preliminary subjective assessment with six strong amateur Go players (1-kyu to 6-dan) from Japan and Taiwan. We targeted strong amateurs because we expected that such players could concretely explain their evaluations, which can be further utilized in improvements. We asked the participants to compare **preaz_2d** and its PAMP+**SX** variant. From Fox Go 2d games, we sampled states where the two policies differed a lot (Jensen-Shannon divergence [4] > 0.45 ; see Fig. 4a). Fig. 4b shows an example policy pair. Although such states were rare, we used them to ensure that differences, if any, would be noticeable.

Each participant assessed 20 policy pairs: 10 common to all participants to examine inter-rater variance, and 10 unique to broaden coverage. Within each pair, the policies were randomly ordered and presented blind. For each pair, participants selected the policy they judged more representative of 2-dan human plays (or chose “unable to judge”) and could optionally provide comments.

Table 4: Qualitative evaluation results when separating into the opening, middle game, and endgame phases (defined in this paper based on move numbers).

	#Assessments	HumanSL	Unable to Judge	PAPM+SX
Opening (1–50)	19	9	2	8
Middle (51–150)	42	10	1	31
Endgame (151–)	59	20	6	33

Among the 120 assessments, 72 (60.0%) favored PAPM+SX, 39 (32.5%) favored the original HumanSL, and 9 (7.5%) were “unable to judge.” These results suggest that PAPM+SX was generally perceived as more 2-dan-human-like than `preaz_2d`. Comments in favor of PAPM+SX often cited its handling of finishing local battles. Conversely, comments in favor of the original HumanSL often mentioned that “local battles had finished, so other areas should be considered” or that “in the opening, more flattened probability distributions appear natural.”

Table 4 further breaks down the assessments into the opening, middle game, and endgame phases (defined as 1–50, 51–150, and 151– moves in this paper). Although the number of opening-phase assessments was relatively small, the results suggested that the original HumanSL may appear more natural in this phase, which is consistent with the participants’ comments. In the future, it would be valuable to analyze other aspects beyond game phases (e.g., life-and-death situations, urgent points, and *ko*) to further enhance human-likeness.

6 Conclusions and Future Work

In this paper, we examined how to utilize KataGo’s HumanSL model as a human-like opponent and proposed to adjust the predicted policies (i.e., probability distributions of moves) based on the distance to the previous move, favoring nearer ones. The design of Policy Adjustment based on Previous Move (PAPM) was motivated by the observation that amateur players, especially at lower skill levels, often respond to the opponent’s most recent move locally rather than considering distant areas of the board.

We compared three selection methods of policies, greedy (i.e., selecting the maximum-probability move, $p' \propto p^\infty$), direct sampling (i.e., $p' \propto p^1$), and softmax interpolation (i.e., $p' \propto p^\rho$ for $\rho \in (1, \infty)$). The results showed that greedy tended to be overly strong and sampling tended to be slightly weak, while softmax allowed tuning of skill levels by adjusting the parameter ρ . Quantitative evaluations using move-matching accuracy, likelihood, and distance to the previous move showed that PAPM with softmax (PAPM+SX) improved human-likeness, particularly for weaker players. A preliminary qualitative assessment, conducted with six strong amateurs based on game states from 2-dan human play, also favored PAPM+SX compared to the original HumanSL.

Future work includes improving the model and expanding analysis. For model improvement, we plan to investigate filtering strategies to avoid selecting moves with extremely low probabilities (e.g., 0.001), integrate PAPM+SX with the

blend method [7,8], and develop automated parameter tuning. For analysis, promising directions include examining why HumanSL fails to reproduce certain human-like tendencies, evaluating PAMP+SX in professional-level games, and designing new human-likeness metrics (e.g., separating distance into dx and dy , or counting captured stones). We also aim to explore other human-like behaviors, such as resignation timing and tactical preferences.

Finally, we plan to conduct further human evaluations involving more participants and diverse datasets to assess how natural the generated policies appear to players. If human-like policies can be developed, they could be applied to teaching games for amateur players [2]. Moreover, with policies of different amateur levels, identifying gaps between specific levels (e.g., 3k vs 1d) may provide insights for instructional strategies.

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⁸ <https://chatgpt.com/>